

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			Indicative content: Components: Connect an ammeter in series to measure the current through the lamp. Connect a voltmeter in parallel with the lamp to measure the voltage across it. Method: 1. Adjust the variable resistor until the voltmeter reads e.g. 1 V. 2. Record the readings of voltage and current. 3. Repeat steps 1 to 2, increasing the voltage by e.g. 1 V each time, until the voltmeter reads 12 V or until the full range of the variable resistor has been used. 5–6 marks Detailed description of both areas. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3–4 marks Detailed description of one areas or limited descriptions of both. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i>	6			6		6

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1–2 marks A limited description of one area. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)			Equal increases in voltage give smaller increases in current or current is increasing at a decreasing rate or the gradient is decreasing (1) showing an increase in resistance (1) [so disagree] N.B. this mark is a follow on one from the 1 st mark			2	2		2
				Question 2 total	6	0	2	8	0	8